

South Indian Movie Box Office & Revenue Intelligence Dashboard

Data Quality & Cleaning Report

1. Dataset Overview

Metric	Result
Records	560
Original columns	9
Final columns	15
Duplicate records	0
Genres	5
Invalid IMDb records	1
Missing Trailer Views	1

2. Cleaning Methodology

- Column names were standardized to lowercase, underscore-separated names.
- Trailer Views were converted to numeric values; the Jersey record remained missing because no defensible value was available.
- IMDb ratings were validated against the 0–10 range. Maya's value of 43.0 was treated as invalid and converted to missing rather than guessing a replacement.
- No duplicate records were removed because no duplicates were found.
- Zero box-office revenue was retained and flagged rather than deleted.
- Very low budgets were retained because unusual values can represent legitimate low-budget films and materially affect ROI.
- Director and lead-actor success-rate values of zero were retained and separately flagged.
- Profit was calculated as Box Office Revenue minus Budget.
- ROI was calculated as Profit divided by Budget, expressed as a percentage.
- Hit/Flop was defined using profitability: Profit > 0 Cr = Hit; Profit <= 0 Cr = Flop.

3. Data Quality Findings

Issue	Finding	Action
Duplicate records	0	No action required
Invalid IMDb	Maya = 43.0	Converted to NaN
Missing Trailer Views	Jersey	Retained as NaN
Zero revenue	1 movie	Retained and flagged
Missing original values	None	No imputation required

4. Genre Distribution

Genre	Movie Count
Romance	145
Action	129
Comedy	124
Thriller	107
Drama	55

5. Profitability Analysis

Statistic	Profit (Cr)
Mean	45.10
Minimum	-582.00
25th percentile	-3.88
Median	7.50
75th percentile	37.25
Maximum	1926.00

Using the project business rule, 374 movies are classified as Hits and 186 as Flops.

6. ROI Analysis

Statistic	ROI (%)
Mean	542.55
Minimum	-100.00
25th percentile	-33.08
Median	53.77
75th percentile	229.00
Maximum	179900.00

Interpretation: ROI is highly right-skewed. The mean ROI is much higher than the median because very small budgets can generate extremely large percentage returns. These observations should be handled carefully in dashboard visuals and business interpretation rather than removed as outliers.

7. Revenue & Quality Status

Status	Count
Revenue available	559
Zero revenue	1
Valid IMDb	559
Missing IMDb	1
Valid Trailer Views	559
Missing Trailer Views	1

8. Final Conclusion

The dataset contains 560 movie records and was transformed into a 15-column analytical dataset. No duplicate records were present. Two data-quality exceptions were retained transparently: Maya's invalid IMDb value was converted to missing, and Jersey's unavailable Trailer Views value remains missing. Financial measures including Profit and ROI were derived consistently from Budget and Box Office Revenue. The resulting dataset is ready for the next project stage: SQL analysis.